## CS4308 Graph Machine Learning

# Fifty papers to read and build

A curated path from graph structure and random walks to neural message passing, transformers, generation, and reliable evaluation. The selection combines the instructor's graph-learning research with influential methods and evaluation frameworks from the wider field. Bibliographic sources were checked in November 2026. The year shown is the cited publication year; an arXiv preprint may be earlier. Workshop papers are identified explicitly.

## How to use this list

The class collectively discusses all 50 papers. Each group, normally a pair, must understand, implement, evaluate, and present at least five distinct papers. Register a balanced selection: one random-walk or spectral method, one message-passing or attention method, one advanced method, one evaluation or reliability paper, and one project-aligned paper. For theory and benchmark papers, implement a central diagnostic or evaluation protocol.

Submit a critical brief, reproducible code, a central result, one ablation or stress test, and a presentation for each study. Small-scale reproductions are acceptable when explicitly scoped. Both partners must contribute and defend the work. Suggested checkpoints are weeks 4, 6, 9, 11, and 13. This reading companion uses the same P01–P50 identifiers as the course outline and website.

GAN means generative adversarial network; GAT means graph attention network. The original GAN paper supplies the general prerequisite, followed by GraphGAN and MolGAN for graph-based adversarial learning.

## Weekly reading map

**Week 2: Spectral structure and graph descriptors.** P01, P02, P06

**Week 3: Walk embeddings and statistical representations.** P03, P04, P05, P07, P08

**Week 4: From graph signals to GCNs.** P12, P13, P16

**Week 5: Message passing and attention.** P14, P15, P17, P20

**Week 6: Graph prediction and feature construction.** P10, P18, P19, P21, P22

**Week 7: Expressivity, failure modes and evaluation.** P24, P25, P26, P48

**Week 9: Scaling and self-supervision.** P09, P11, P27, P28, P29, P30, P31

**Week 10: Graph learning in practice.** P32, P33, P34, P35, P36, P37, P38, P39

**Week 11: Graph transformers and structural encodings.** P23, P40, P41

**Week 12: Adversarial and generative graph learning.** P42, P43, P44, P45

**Week 13: Trustworthy graph research.** P46, P47, P49, P50

## Selection and coverage

The 50-paper set includes 22 publications coauthored by Mudassir Shabbir on graph learning and its representation foundations, plus 28 influential papers selected for their significance to the field.

Author coverage comes first: include the publicly verifiable graph-learning papers, graph descriptors, graph-learning applications, and directly relevant graph-distinguishability foundations coauthored by Mudassir Shabbir. Coverage was checked on 2 November 2026; this does not assert coverage of unpublished or unlisted manuscripts.

The remaining 28 readings are chosen for foundational ideas, influential model families, and established benchmarks or evaluation methods. This is a reasoned syllabus selection, not a citation-count ranking. It preserves PageRank, DeepWalk, node2vec, GCN, GAN, GAT and graph transformers, with message passing, self-supervision, and trustworthy evaluation.

A preprint and its published version count as one reading. Distinct conference and extended journal papers are retained where they make separate contributions; their relationship is noted. Preprints and works in progress are labeled. General geometry and control papers without a direct graph-learning contribution are outside this reading set.

All 50 receive class coverage through lectures, concise paper spotlights and discussions. The assigned week identifies the main discussion; groups may choose papers across weeks for their five-paper portfolio.

Coverage sources: author ORCID; coauthor publication list; coverage and version audit. Each entry links its primary publication or manuscript.

## Foundations and representations

### P01 The PageRank Citation Ranking: Bringing Order to the Web

Lawrence Page et al. Stanford InfoLab Technical Report 1999-66, 1999.

FIELD LANDMARK

**Why read it.** Connects graph centrality, Markov chains, and ranking before learned representations.

**Build or investigate.** Implement power iteration with teleportation; test dangling nodes and sensitivity to the damping factor.

**Level.** Foundation Primary source ↗

### P02 Laplacian Eigenmaps and Spectral Techniques for Embedding and Clustering

Mikhail Belkin and Partha Niyogi. NeurIPS (then NIPS), 2001.

FIELD LANDMARK

**Why read it.** Introduces the graph Laplacian as a bridge between geometry, embeddings, and spectral learning.

**Build or investigate.** Compute two-dimensional spectral embeddings of a small graph; compare neighborhoods with a random layout.

**Level.** Foundation Primary source ↗

### P03 DeepWalk: Online Learning of Social Representations

Bryan Perozzi et al. KDD, 2014.

FIELD LANDMARK

**Why read it.** Turns short random walks into a practical unsupervised representation-learning pipeline.

**Build or investigate.** Implement walk generation and skip-gram training; evaluate node classification with a fixed split.

**Level.** Foundation Primary source ↗

### P04 node2vec: Scalable Feature Learning for Networks

Aditya Grover and Jure Leskovec. KDD, 2016.

FIELD LANDMARK

**Why read it.** Shows how biased walks trade off local neighborhoods and structural roles.

**Build or investigate.** Sweep return and exploration parameters against DeepWalk using identical downstream evaluation.

**Level.** Foundation Primary source ↗

### P05 Estimating Descriptors for Large Graphs

Zohair Raza Hassan, Mudassir Shabbir, Imdadullah Khan, Waseem Abbas. PAKDD 2020, LNCS 12084, 779-791, 2020.

MUDASSIR SHABBIR / RESEARCH

Conference paper; the extended edge-stream journal paper is a separate reading.

**Why read it.** Streaming graph descriptors for classification.

**Build or investigate.** Implement reservoir-sampled graphlet counts; plot descriptor error and classification accuracy against the memory budget.

**Level.** Intermediate Primary source ↗

### P06 Interpretable Multi-Scale Graph Descriptors via Structural Compression

Ammar Ahmed, Zohair Raza Hassan, Mudassir Shabbir. Information Sciences 533, 169-180, 2020.
MUDASSIR SHABBIR / RESEARCH
**Why read it.** Interpretable multiscale graph representations for classification via structural compression.
**Build or investigate.** Implement structural compression at several scales; compare descriptor discrimination and runtime with degree and spectral summaries.
**Level.** Intermediate Primary source ↗

### P07 DGSD: Distributed Graph Representation via Graph Statistical Properties

Anwar Said, Saeed-Ul Hassan, Suppawong Tuarob, Raheel Nawaz, Mudassir Shabbir. Future Generation Computer Systems 119, 166-175, 2021.
MUDASSIR SHABBIR / RESEARCH
**Why read it.** Distributed permutation-invariant graph embedding.
**Build or investigate.** Construct statistical descriptors under different graph partitions; test permutation invariance and compare with a degree-histogram baseline.
**Level.** Intermediate Primary source ↗

### P08 NetKI: A Kirchhoff Index Based Statistical Graph Embedding in Nearly Linear Time

Anwar Said, Saeed-Ul Hassan, Waseem Abbas, Mudassir Shabbir. Neurocomputing 433, 108-118, 2021.
MUDASSIR SHABBIR / RESEARCH
**Why read it.** Scalable graph representation and graph classification using effective resistance.
**Build or investigate.** Compute exact effective resistances on small graphs, compare an approximate NetKI representation, and measure runtime versus classification quality.
**Level.** Intermediate Primary source ↗

### P09 Computing Graph Descriptors on Edge Streams

Zohair Raza Hassan, Sarwan Ali, Imdadullah Khan, Mudassir Shabbir, Waseem Abbas. ACM Transactions on Knowledge Discovery from Data 17(8), article 112, 2023.
MUDASSIR SHABBIR / RESEARCH
Journal extension of the 2020 descriptor paper; reproduce its additional contribution.
**Why read it.** Streaming structural/spectral graph descriptors and graph classification.
**Build or investigate.** Reproduce one streaming descriptor and compare against exact computation across memory budgets and stream orderings; isolate the journal extension.
**Level.** Intermediate Primary source ↗

### P10 On Augmenting Topological Graph Representations for Attributed Graphs

Anwar Said, Mudassir Shabbir, Saeed-Ul Hassan, Zohair Raza Hassan, Ammar Ahmed, Xenofon Koutsoukos. Applied Soft Computing 136, 110104, 2023.
MUDASSIR SHABBIR / RESEARCH
**Why read it.** Integrating graph attributes with topological representations for graph classification.
**Build or investigate.** Compare topology-only, attributes-only and augmented representations under fixed splits to test when node attributes add useful information.
**Level.** Intermediate Primary source ↗

### P11 MSDGSD: A Scalable Graph Descriptor for Processing Large Graphs

Muhammad Ali, Anwar Said, Iqra Safder, Saeed Ul Hassan, Naif Radi Aljohani, Mudassir Shabbir. IEEE Transactions on Computational Social Systems 11(3), 3594-3605, 2024.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Scalable graph descriptors for downstream machine learning.

**Build or investigate.** Implement a tractable multiscale descriptor; compare DGSD and MSDGSD at matched compute and measure accuracy and runtime as graph size grows.

**Level.** Intermediate Primary source ↗

## Graph neural networks

### P12 The Graph Neural Network Model

Franco Scarselli, Marco Gori, Ah Chung Tsoi, Markus Hagenbuchner, Gabriele Monfardini. IEEE Transactions on Neural Networks 20(1), 61-80, 2009.

FIELD LANDMARK

**Why read it.** An early general formulation of neural computation on graphs; provides historical grounding for today's message-passing architectures.

**Build or investigate.** Implement a small recurrent graph-state update and examine convergence; compare its assumptions with a fixed-depth GCN.

**Level.** Advanced Primary source ↗

### P13 Convolutional Neural Networks on Graphs with Fast Localized Spectral Filtering

Michaël Defferrard et al. NeurIPS (then NIPS), 2016.

FIELD LANDMARK

**Why read it.** Replaces expensive eigendecomposition with localized polynomial filters.

**Build or investigate.** Implement Chebyshev recurrence; compare polynomial order, receptive field, runtime, and accuracy.

**Level.** Intermediate Primary source ↗

### P14 Inductive Representation Learning on Large Graphs

William L. Hamilton et al. NeurIPS (then NIPS), 2017.

FIELD LANDMARK

**Why read it.** Introduces neighborhood sampling and aggregation for generalization to unseen nodes.

**Build or investigate.** Implement mean aggregation with sampling and evaluate a genuine inductive node split.

**Level.** Intermediate Primary source ↗

### P15 Neural Message Passing for Quantum Chemistry

Justin Gilmer et al. ICML, 2017.

FIELD LANDMARK

**Why read it.** Unifies many graph architectures through message, update, and readout functions.

**Build or investigate.** Build a small edge-aware message-passing model for one molecular regression target.

**Level.** Intermediate Primary source ↗

### P16 Semi-Supervised Classification with Graph Convolutional Networks

Thomas N. Kipf and Max Welling. ICLR, 2017.

FIELD LANDMARK

**Why read it.** Establishes a compact, reproducible baseline for learning from node features and edges.

**Build or investigate.** Implement a two-layer GCN; ablate self-loops, normalization, and feature-only baselines.

**Level.** Foundation Primary source ↗

### P17 Graph Attention Networks

Petar Veličković et al. ICLR, 2018.

FIELD LANDMARK

**Why read it.** Introduces learned neighborhood weighting and multi-head attention.

**Build or investigate.** Implement masked graph attention; compare heads and attention dropout with GCN.

**Level.** Intermediate Primary source ↗

### P18 Hierarchical Graph Representation Learning with Differentiable Pooling

Rex Ying et al. NeurIPS, 2018.

FIELD LANDMARK

**Why read it.** Introduces learned graph coarsening for prediction over entire graphs.

**Build or investigate.** Implement one DiffPool stage and compare it with global mean pooling under matched budgets.

**Level.** Intermediate Primary source ↗

### P19 How Powerful are Graph Neural Networks?

Keyulu Xu et al. ICLR, 2019.

FIELD LANDMARK

**Why read it.** Relates aggregation to graph isomorphism and motivates the Graph Isomorphism Network.

**Build or investigate.** Implement GIN and test sum versus mean aggregation on controlled multiset and graph examples.

**Level.** Advanced Primary source ↗

### P20 Enhanced Graph Neural Networks with Ego-Centric Spectral Subgraph Embeddings Augmentation

Anwar Said, Mudassir Shabbir, Tyler Derr, Waseem Abbas, Xenofon Koutsoukos. 22nd IEEE International Conference on Machine Learning and Applications (ICMLA), 2023.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Spectral ego-subgraph embeddings augment GNN node features.

**Build or investigate.** Augment a GCN with ego-subgraph spectral features; vary neighborhood radius and include feature-only and random-feature controls.

**Level.** Intermediate Primary source ↗

### P21 Improving Graph Machine Learning Performance Through Feature Augmentation Based on Network Control Theory

Anwar Said, Obaid Ullah Ahmad, Waseem Abbas, Mudassir Shabbir, Xenofon Koutsoukos. 32nd Mediterranean Conference on Control and Automation (MED), 322-327, 2024.

MUDASSIR SHABBIR / RESEARCH

Conference study; compare with the expanded 2025 rank-encoding paper.

**Why read it.** Network control based node-feature augmentation for GNNs.

**Build or investigate.** Add network-control features to a fixed GNN; compare no features, random features and spectral features while holding the training budget constant.

**Level.** Intermediate Primary source ↗

### P22 Network Controllability Perspectives on Graph Representation

Anwar Said, Obaid Ullah Ahmad, Waseem Abbas, Mudassir Shabbir, Xenofon Koutsoukos. IEEE Transactions on Knowledge and Data Engineering 36(8), 4116-4128, 2024.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Control-theoretic graph representation for machine learning.

**Build or investigate.** Construct a control-theoretic graph representation and compare it with spectral and degree features under the same classifier and split.

**Level.** Advanced Primary source ↗

### P23 Feature Construction Using Network Control Theory and Rank Encoding for Graph Machine Learning

Anwar Said, Yifan Wei, Obaid Ullah Ahmad, Mudassir Shabbir, Waseem Abbas, Xenofon Koutsoukos. IEEE Open Journal of Control Systems, 2025.

MUDASSIR SHABBIR / RESEARCH

Expanded journal study following the 2024 feature-augmentation paper.

**Why read it.** Controllability and rank encoding generate expressive node features for graph classification.

**Build or investigate.** Reproduce rank-based node features on small benchmarks; compare the 2024 feature construction, rank encoding and standard positional encodings.

**Level.** Advanced Primary source ↗

## Expressivity and limits

### P24 Beyond Homophily in Graph Neural Networks: Current Limitations and Effective Designs

Jiong Zhu et al. NeurIPS, 2020.

FIELD LANDMARK

**Why read it.** Studies graphs whose connected nodes differ and motivates separating ego and neighbor information.

**Build or investigate.** Implement an H2GCN-style baseline and vary homophily on synthetic graphs.

**Level.** Intermediate Primary source ↗

### P25 On the Bottleneck of Graph Neural Networks and its Practical Implications

Uri Alon and Eran Yahav. ICLR, 2021.

FIELD LANDMARK

**Why read it.** Introduces over-squashing as a failure of information transmission across graph bottlenecks.

**Build or investigate.** Reproduce a small long-range task and compare added shortcuts with increased hidden dimension.

**Level.** Advanced Primary source ↗

### P26 Control-based Conditions for Graph Distinguishability

Muhammad Raza, Obaid Ullah Ahmad, Mudassir Shabbir, Xenofon Koutsoukos, Waseem Abbas. SIAM Conference on Control and its Applications (CT), 31-38, 2025.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Uses control-theoretic conditions to study graph distinguishability, connecting spectral structure with the limits of graph representations.

**Build or investigate.** Enumerate small graph pairs and test walk-matrix conditions alongside spectral and 1-WL signatures; report what each criterion can and cannot distinguish.

**Level.** Advanced Primary source ↗

## Scaling and self-supervision

### P27 Deep Graph Infomax

Petar Veličković et al. ICLR, 2019.

FIELD LANDMARK

**Why read it.** Introduces node representation learning through local-global contrast.

**Build or investigate.** Implement DGI and test feature corruption with a frozen linear evaluation protocol.

**Level.** Intermediate Primary source ↗

### P28 Graph Contrastive Learning with Augmentations

Yuning You et al. NeurIPS, 2020.

FIELD LANDMARK

**Why read it.** Makes graph augmentations a central design choice in contrastive pretraining.

**Build or investigate.** Implement two graph augmentations and assess whether they preserve labels on the chosen dataset.

**Level.** Intermediate Primary source ↗

### P29 GraphSAINT: Graph Sampling Based Inductive Learning Method

Hanqing Zeng et al. ICLR, 2020.

FIELD LANDMARK

**Why read it.** Studies subgraph sampling with normalization to improve training efficiency.

**Build or investigate.** Implement one subgraph sampler and test the contribution of sampling normalization.

**Level.** Intermediate Primary source ↗

### P30 Control-Based Graph Embeddings with Data Augmentation for Contrastive Learning

Obaid Ullah Ahmad, Anwar Said, Mudassir Shabbir, Xenofon Koutsoukos, Waseem Abbas. American Control Conference (ACC), 27-32, 2024.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Unsupervised contrastive graph representation with control-informed augmentation.

**Build or investigate.** Compare control-informed graph augmentations with random edge deletion under the same contrastive objective; ablate augmentation strength.

**Level.** Advanced Primary source ↗

### P31 Learning Backbones: Sparsifying Graphs Through Zero Forcing for Effective Graph-Based Learning

Obaid Ullah Ahmad, Anwar Said, Mudassir Shabbir, Xenofon Koutsoukos, Waseem Abbas. Complex Networks & Their Applications XIII (COMPLEX NETWORKS 2024), 85-100, 2025.

MUDASSIR SHABBIR / RESEARCH

Conference held in 2024; proceedings published in 2025. The arXiv version is the same reading.

**Why read it.** Zero-forcing graph sparsification for downstream GNN classification.

**Build or investigate.** Extract zero-forcing backbones; compare with random and degree-based edge retention at matched sparsity, reporting accuracy, runtime and connectivity.

**Level.** Intermediate Primary source ↗

## Applications and graph optimization

### P32 Leveraging Deep Learning and SNA Approaches for Smart City Policing in the Developing World

Saeed-Ul Hassan, Mudassir Shabbir, Sehrish Iqbal, Anwar Said, Faisal Kamiran, Raheel Nawaz, Umar Saif. International Journal of Information Management 56, 102045, 2021.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Direct GCN application to binary node classification in communication networks; suitable critical discussion of privacy, bias and evaluation.

**Build or investigate.** Use synthetic communication graphs to study label prediction and structural bias; evaluate leakage and false-positive disparities without profiling real individuals.

**Level.** Intermediate Primary source ↗

### P33 Attack-Resilient Multi-Agent Flocking Control Using Graph Neural Networks

Chandreyee Bhowmick, Mudassir Shabbir, Xenofon Koutsoukos. 30th Mediterranean Conference on Control and Automation (MED), 300-305, 2022.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** GCN-based distributed flocking resilient to adversarial communication.

**Build or investigate.** Simulate a small flock under compromised neighbor messages; compare the GNN controller with a nominal controller using collision and cohesion metrics.

**Level.** Intermediate Primary source ↗

### P34 Circuit Design Completion Using Graph Neural Networks

Anwar Said, Mudassir Shabbir, Brian Broll, Waseem Abbas, Peter Völgyesi, Xenofon Koutsoukos. Neural Computing and Applications 35, 12145-12157, 2023.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** GNNs for missing circuit components and connections.

**Build or investigate.** Mask a component or connection in small circuit graphs; compare GNN completion with frequency and topology baselines using circuit-level held-out splits.

**Level.** Intermediate Primary source ↗

### P35 Learning-Based Heuristic for Combinatorial Optimization of the Minimum Dominating Set Problem using Graph Convolutional Networks

Abihith Kothapalli, Mudassir Shabbir, Xenofon Koutsoukos. arXiv:2306.03434, 2023.

MUDASSIR SHABBIR / RESEARCH

Preprint.

**Why read it.** GCNs for NP-hard graph combinatorial optimization.

**Build or investigate.** Train on small synthetic graphs; compare feasible dominating sets against greedy and exact solutions, then test size and graph-family generalization.

**Level.** Intermediate Primary source ↗

### P36 Sequential Graph Neural Networks for Source Code Vulnerability Identification

Ammar Ahmed, Anwar Said, Mudassir Shabbir, Xenofon Koutsoukos. HotSoS 2023 work in progress; arXiv:2306.05375, 2023.

MUDASSIR SHABBIR / RESEARCH

Work-in-progress/preprint version; no journal publication is claimed.

**Why read it.** Sequential/GNN code representation, vulnerability classification and CVEFGE dataset.

**Build or investigate.** Build a small code-graph pipeline and compare GNN versus sequence-only features; split by project and remove near-duplicate functions before evaluation.

**Level.** Intermediate Primary source ↗

### P37 A Graph Machine Learning Framework to Compute Zero Forcing Sets in Graphs

Obaid Ullah Ahmad, Mudassir Shabbir, Waseem Abbas, Xenofon Koutsoukos. IEEE Transactions on Network Science and Engineering 11(2), 2110-2123, 2024.

MUDASSIR SHABBIR / RESEARCH

**Why read it.** Graph ML solution to zero-forcing set computation.

**Build or investigate.** Train a graph learner to propose zero-forcing sets; verify feasibility by simulation and compare set size and runtime with greedy and exact small-graph solvers.

**Level.** Advanced Primary source ↗

## Relational and temporal learning

### P38 Modeling Relational Data with Graph Convolutional Networks

Michael Schlichtkrull et al. ESWC, 2018.

FIELD LANDMARK

**Why read it.** Extends message passing to typed relations and knowledge-graph completion.

**Build or investigate.** Implement basis-decomposed R-GCN on a small relational dataset with leakage-aware edge splits.

**Level.** Intermediate Primary source ↗

### P39 Inductive Representation Learning on Temporal Graphs

Da Xu et al. ICLR, 2020.

FIELD LANDMARK

**Why read it.** Introduces temporal attention with continuous-time encodings.

**Build or investigate.** Implement TGAT-style temporal neighbors and verify every prediction uses only past events.

**Level.** Advanced Primary source ↗

## Graph transformers

### P40 Do Transformers Really Perform Bad for Graph Representation?

Chengxuan Ying et al. NeurIPS, 2021.
FIELD LANDMARK
**Why read it.** Shows how structural encodings make global attention useful for graph prediction.
**Build or investigate.** Implement a compact Graphormer and ablate degree and shortest-path encodings.
**Level.** Advanced Primary source ↗

### P41 Recipe for a General, Powerful, Scalable Graph Transformer

Ladislav Rampášek et al. NeurIPS, 2022.
FIELD LANDMARK
**Why read it.** Combines local message passing, global attention, and position or structure encodings.
**Build or investigate.** Implement one GPS block and perform a three-component ablation under a fixed parameter budget.
**Level.** Advanced Primary source ↗

## Generative graph learning

### P42 Generative Adversarial Nets

Ian J. Goodfellow et al. NeurIPS (then NIPS), 2014.
FIELD LANDMARK
**Why read it.** Provides the adversarial-learning prerequisite; this paper itself is not a graph model.
**Build or investigate.** Implement a small GAN on a toy distribution and diagnose instability before attempting graph generation.
**Level.** Foundation Primary source ↗

### P43 GraphGAN: Graph Representation Learning with Generative Adversarial Nets

Hongwei Wang et al. AAAI, 2018.
FIELD LANDMARK
**Why read it.** Applies adversarial training to connectivity distributions for graph embeddings.
**Build or investigate.** Implement a small generator-discriminator link model and compare with a random-walk embedding baseline.
**Level.** Advanced Primary source ↗

### P44 GraphRNN: Generating Realistic Graphs with Deep Auto-regressive Models

Jiaxuan You et al. ICML, 2018.
FIELD LANDMARK
**Why read it.** Provides an autoregressive alternative to adversarial graph generation.
**Build or investigate.** Implement a small sequential graph generator and compare degree and clustering distributions.
**Level.** Advanced Primary source ↗

### P45 MolGAN: An implicit generative model for small molecular graphs

Nicola De Cao and Thomas Kipf. ICML Workshop on Theoretical Foundations and Applications of Deep Generative Models, 2018.

FIELD LANDMARK

**Why read it.** Combines adversarial graph generation with property objectives and exposes mode collapse.

**Build or investigate.** Implement a reduced small-molecule generator; report validity, uniqueness, novelty, and diversity.

**Level.** Advanced Primary source ↗

## Evaluation and trust

### P46 Adversarial Attacks on Graph Neural Networks via Meta Learning

Daniel Zügner and Stephan Günnemann. ICLR, 2019.

FIELD LANDMARK

**Why read it.** Studies sensitivity to training-graph perturbations and motivates robust evaluation.

**Build or investigate.** On a public benchmark, compare controlled edge perturbations with random edits and report accuracy degradation.

**Level.** Advanced Primary source ↗

### P47 GNNExplainer: Generating Explanations for Graph Neural Networks

Rex Ying et al. NeurIPS, 2019.

FIELD LANDMARK

**Why read it.** Explores which subgraphs and features support a GNN prediction.

**Build or investigate.** Implement explanation masks on a motif dataset; measure fidelity, sparsity, and stability.

**Level.** Intermediate Primary source ↗

### P48 Open Graph Benchmark: Datasets for Machine Learning on Graphs

Weihua Hu et al. NeurIPS, 2020.

FIELD LANDMARK

**Why read it.** Supplies task-specific datasets, realistic splits, and standardized evaluators.

**Build or investigate.** Implement an OGB baseline and document why its official split and metric match the task.

**Level.** Foundation Primary source ↗

### P49 A Survey of Graph Unlearning

Anwar Said, Ngoc N. Tran, Yuying Zhao, Tyler Derr, Mudassir Shabbir, Waseem Abbas, Xenofon Koutsoukos. arXiv:2310.02164; revised October 2025, 2023.

MUDASSIR SHABBIR / RESEARCH

First posted in 2023; this reading uses the October 2025 revision.

**Why read it.** Survey of graph machine unlearning, privacy, evaluation and applications.

**Build or investigate.** Select one surveyed unlearning method; compare deletion utility, retained-task performance and compute against full retraining using a clearly stated threat model.

**Level.** Intermediate Primary source ↗

**P50 NeuroGraph: Benchmarks for Graph Machine Learning in Brain Connectomics**
Anwar Said, Roza G. Bayrak, Tyler Derr, Mudassir Shabbir, Daniel Moyer, Catie Chang, Xenofon Koutsoukos. NeurIPS 2023 Datasets and Benchmarks, 2023.
MUDASSIR SHABBIR / RESEARCH
**Why read it.** Graph ML benchmark for brain connectomics.
**Build or investigate.** Reproduce a small benchmark with authorized data, subject-disjoint splits and a non-graph baseline; test graph-construction sensitivity without making clinical claims.
**Level.** Intermediate Primary source ↗

## Twenty illustrative research projects

A rigorous replication with a meaningful extension, a careful negative result, or a focused new method can all make a strong project. Publication acceptance is not a grading criterion. The required paper is submitted to the course; external conference or journal submission is optional and requires agreement among the authors.

These are scoped research suggestions. Agree on a dataset, question, and compute budget before beginning. Each project should report competitive and simple baselines, valid splits, ablations, error analysis, and reproducible artifacts.

### R01 When do biased walks help?

Which graph structures make node2vec's walk bias useful?
**Data.** Synthetic stochastic block models and Cora. Dataset documentation
**Baselines.** DeepWalk, node2vec, degree features plus logistic regression
**Evaluation.** Node macro-F1 and accuracy; sweep homophily, walk length, p and q under fixed embedding budgets.
**Extension.** Relate performance to mixing time and community strength.
**Compute.** CPU-friendly

### R02 PageRank as a learning prior

Can personalized diffusion improve classification when labels are scarce?
**Data.** Cora and CiteSeer. Dataset documentation
**Baselines.** MLP, GCN, personalized PageRank propagation
**Evaluation.** Accuracy versus labeled fraction; fix splits and compare propagation depth and damping.
**Extension.** Choose damping using local graph structure without test labels.
**Compute.** CPU or small GPU

### R03 Attention under noisy edges

Does learned graph attention remain useful as irrelevant edges increase?
**Data.** Cora plus controlled random edge additions. Dataset documentation
**Baselines.** GCN, GraphSAGE, GAT
**Evaluation.** Accuracy and calibration versus edge corruption; match parameter counts and tuning budgets.
**Extension.** Compare attention entropy with robustness and test a sparsity regularizer.
**Compute.** Small GPU

### R04 Learning across unlike neighbors

When do neighborhood aggregation models fail on heterophilous graphs?
**Data.** Roman-empire and Minesweeper benchmark splits. Dataset documentation
**Baselines.** MLP, GCN, GAT and a heterophily-aware model
**Evaluation.** Dataset-appropriate accuracy or ROC-AUC; preserve supplied splits and report multiple seeds.
**Extension.** Adapt the balance of self features and neighbor features.
**Compute.** Small GPU

### R05 How deep can a GNN go?

Which changes preserve predictive information as network depth increases?
**Data.** Cora and synthetic community graphs. Dataset documentation
**Baselines.** Plain GCN, residual GCN and normalization variants
**Evaluation.** Accuracy, embedding similarity and Dirichlet energy across depth; control training budget.
**Extension.** Separate optimization failure from oversmoothing with controlled interventions.
**Compute.** CPU or small GPU

### R06 Rewiring graph bottlenecks

Can a few added edges improve long-range information transfer?
**Data.** Synthetic trees, barbell graphs and distance-label tasks. Dataset documentation
**Baselines.** GCN or MPNN with no rewiring, random rewiring and a topology-aware rule
**Evaluation.** Long-range task accuracy, path length and runtime at equal edge budgets.
**Extension.** Study the trade-off between shortcut benefits and structural distortion.
**Compute.** CPU-friendly

### R07 A fair transformer comparison

When does global attention earn its extra compute?
**Data.** ZINC benchmark graphs. Dataset documentation
**Baselines.** GIN, a graph transformer and a local/global hybrid
**Evaluation.** Regression MAE, peak memory and runtime; match parameters and training budget.
**Extension.** Identify graph sizes or structures where global attention helps.
**Compute.** Small GPU; use a declared subset for pilots

### R08 Which positional encoding matters?

How stable are graph transformers to positional encoding choices?
**Data.** ZINC and synthetic symmetric graphs. Dataset documentation
**Baselines.** Transformer without positional encoding, Laplacian encoding and random-walk encoding
**Evaluation.** MAE and permutation consistency; vary signs, eigenspaces and graph size.
**Extension.** Test sign-invariant or noise-robust encoding variants.
**Compute.** Small GPU

### R09 Self-supervision with few labels

Which graph augmentations help when supervision is limited?
**Data.** Cora and CiteSeer. Dataset documentation
**Baselines.** Supervised GCN, DGI and a contrastive graph encoder
**Evaluation.** Linear-probe accuracy across label budgets; tune augmentations on validation data only.
**Extension.** Replace uniform edge dropping with a structure-aware rule.
**Compute.** Small GPU

### R10 Masked graph reconstruction

Can reconstructing masked features improve robustness to missing attributes?
**Data.** Cora and PubMed. Dataset documentation
**Baselines.** MLP, GCN and GraphMAE-style pretraining
**Evaluation.** Classification accuracy and reconstruction loss across random and structured missingness.
**Extension.** Compare degree-aware masking with uniform masking.
**Compute.** Small GPU

### R11 Predicting future collaborations

How much do graph embeddings improve over simple link heuristics?
**Data.** ogbl-collab. Dataset documentation
**Baselines.** Common neighbors, Adamic-Adar, node2vec and GraphSAGE
**Evaluation.** Official Hits@50 with official temporal splits and negative candidates; document allowed edges.
**Extension.** Add recency-aware edge weights and audit temporal leakage.
**Compute.** Moderate GPU; prototype on a declared subset

### R12 Learning from event streams

When does temporal memory beat a static graph encoder?
**Data.** Wikipedia temporal interaction data distributed with TGN. Dataset documentation
**Baselines.** Static GraphSAGE, a recency heuristic and TGN
**Evaluation.** Average precision and ROC-AUC with time-ordered splits and explicit negative sampling.
**Extension.** Test memory resets, time encodings and unseen-node performance.
**Compute.** Small GPU

### R13 Relations in a knowledge graph

Do relation-aware messages improve completion beyond embedding models?
**Data.** FB15k-237. Dataset documentation
**Baselines.** DistMult, TransE and R-GCN with a common scoring head where appropriate
**Evaluation.** Filtered MRR and Hits@10; prevent inverse-relation leakage and report relation-wise errors.
**Extension.** Allocate relation parameters by frequency or sparsity.
**Compute.** Small to moderate GPU

### R14 Graph recommendations beyond popularity

Can graph propagation improve recommendation for low-activity users?
**Data.** MovieLens 100K stable benchmark. Dataset documentation
**Baselines.** Popularity, matrix factorization and LightGCN
**Evaluation.** Recall@20 and NDCG@20 on a time-ordered split; keep candidate sampling fixed.
**Extension.** Analyze accuracy and coverage across user-activity groups without inferring sensitive attributes.
**Compute.** CPU or small GPU

### R15 Molecular learning under distribution shift

Do stronger graph encoders generalize to new molecular scaffolds?
**Data.** ogbg-molhiv. Dataset documentation
**Baselines.** Fingerprint classifier, GCN and GIN
**Evaluation.** Official ROC-AUC on scaffold splits; add calibration and performance by scaffold frequency.
**Extension.** Compare pretraining or edge-feature treatments under equal tuning budgets.
**Compute.** Small GPU

### R16 Generating diverse small graphs

Which generative model best preserves structure without memorization?
**Data.** Synthetic community and grid graphs. Dataset documentation
**Baselines.** Erdos-Renyi or block-model generator, GraphRNN and a small graph GAN
**Evaluation.** Validity, uniqueness, novelty, and degree/clustering distribution distances on held-out graphs.
**Extension.** Diagnose mode collapse as graph size and training diversity change.
**Compute.** Small GPU

### R17 Adversarial versus diffusion generation

How do adversarial and diffusion objectives trade quality for cost?
**Data.** Small synthetic graphs; optional QM9 extension. Dataset documentation
**Baselines.** A MolGAN-inspired discrete generator and a small DiGress-style model
**Evaluation.** Validity, diversity, novelty, structural fidelity and training/sample time at matched graph sizes.
**Extension.** If using molecules, add chemical-validity checks and compare conditional generation.
**Compute.** Moderate GPU; synthetic scope recommended

### R18 Are graph explanations stable?

Does a model explanation persist across seeds and harmless perturbations?
**Data.** Synthetic motif graphs with known generating motifs. Dataset documentation
**Baselines.** Gradient saliency, random edge masks and GNNExplainer
**Evaluation.** Fidelity, sparsity, motif precision/recall and agreement across repeated runs.
**Extension.** Build a stability-aware explanation selection rule; distinguish explanation from causality.
**Compute.** CPU or small GPU

### R19 Robustness on a fixed edge budget

Which defenses withstand targeted structural perturbations?
**Data.** Cora or CiteSeer public benchmark graphs. Dataset documentation
**Baselines.** GCN, GAT, edge-drop training and a simple graph denoiser
**Evaluation.** Clean and perturbed accuracy across equal budgets; distinguish random from targeted perturbations.
**Extension.** Test whether defense gains persist on unseen perturbation strategies in an offline benchmark.
**Compute.** Small GPU

### R20 Do benchmark rankings survive protocol changes?

How sensitive are model rankings to splits, tuning and random seeds?
**Data.** Cora plus ogbn-arxiv, with a smaller scoped pilot if needed. Dataset documentation
**Baselines.** MLP, GCN, GraphSAGE and GAT
**Evaluation.** Accuracy distributions, rank stability, runtime and tuning cost; preserve official test protocols.
**Extension.** Measure how small search budgets alter conclusions and publish a reproducibility checklist.
**Compute.** Small to moderate GPU